INCH-POUND

MIL-G-49312(CR)

6 November 1989

MILITARY SPECIFICATION
GOGGLES, NIGHT VISION AN/PVS-7A

This specification is approved for use by USACECOM, Department of Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the AN/PVS-7A, Night Vision Goggle.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards. The following specifications, standards form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-11268	-	Parts, Materials, and Processes Used in Electronic Equipment
MIL-O-13830	-	Optical Components for Fire Control Instruments, Electronic Equipment
MIL-B-49030/6	-	Battery, Dry BA-3058/U
MIL-B-49430/4	-	Battery, Primary Lithium, Sulfur-Dioxide BA-5567/U

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-ED-TO, Fort Monmouth, NJ 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A: FSC 5855

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STANDARDS

MILITARY

MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-150	-	Photographic Lenses
MIL-STD-171	-	Finishing of Metal and Wood Surfaces
MIL-STD-454	-	Standard General Requirements for Electrical Component Parts
MIL-STD-461	-	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-810	-	Environmental Test Methods

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099).

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

A3140630 - Night Vision Goggle, AN/PVS-7A
A3140760 - Goggle Assembly
A3140690 - Facemask Assembly
A3140660 - Case, Carrying Assembly
A3140670 - Case, Shipping and Storage
A3140650 - Filter, Sacrificial
A3140653 - Shield, Demist
A3140850 - Lens Assembly, Objective
A3140766 - Lens Assembly, Binocular Eyepiece
A3140806 - Light Emitting Diode

(Copies of specifications, standard drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. The AN/PVS-7A Night Vision Goggle is a lightweight, head-mounted, self-contained Night Vision System. It is used in starlight and moonlight by individual soldiers for walking, driving vehicles, weapon firing, short-range surveillance, and such manual tasks as map reading, vehicle maintenance, and administering medical aid. Each device consists of an objective lens, image intensifier tube assembly, and eyepiece assembly. The units are affixed to a face mask that is held by head straps to the user's head. The assembly incorporates an IR-emitting light source that provides illumination when required, for close-up viewing. Eyepiece diopter adjustment is provided so the device can be worn with or without corrective eyeglasses.

3.1.1 Interface definition.

3.1.1.1 NVG design. The NVG shall consist of a head-mounted imaging assembly with integral battery pack, carrying case, shipping/storage case, and ancillary equipment. The head-mounted equipment shall consist of a face mask and goggle assembly. Each goggle assembly shall consist of an objective lens, image intensifier tube assembly, and eyepiece assembly. The NVG shall also contain an IR light source, LED indicator, battery pack, and electrical control switch. Ancillary equipment will include antifogging protection and a sacrificial filter for sand protection. The carrying case shall provide protection for the NVG against the conditions associated with carrying the NVG in a field environment. The shipping case provides protection for the NVG against damage associated with transportation, handling, and storage. The face mask shall be a full mask with removable chin strap. The face mask shall be equipped with facepads, which allow the operator to use the system for a 12-hour period without irritation to the skin. The latch mounting assembly shall permit mounting and demounting of the system using one hand. The control switch for the system shall be located on the upper left edge so an infantryman with a rifle in the right hand can operate the switch while wearing arctic gloves or mittens. The LED shall be located on the lower left front of the system to prevent interference of the projected beam when focusing the system with the right hand. In addition, there shall be no interference of the projected beam of the LED by any part of the system. A disconnect circuit shall turn the system OFF when the system is removed from the mask. A separate mode shall permit the system to be used out of the face mask using the same switch sequence as when in the face mask.

3.1.1.2 Helmet interface. The NVG shall be designed for use with the Combat Vehicle Crewmans (CVC), M-1 (Steel), or DH-145 (Kevlar) protective helmets without affecting the protective qualities of the helmet.

3.1.1.3 Protective mask interface. The NVG shall be designed for use with the M17A2, M24, M40, M25A1, XM40, XM41, XM42, and XM43 protective masks without affecting the protective qualities of the protective masks.

3.2 Construction. The assembly shall be in accordance with drawing A3140630 and all provisions of this specification.

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3.3 Qualification. Goggles furnished under this specification shall be product which are authorized by the Government as qualified products (see 3.9.3).

3.3.1 Initial production test. When specified in the contract or purchase order the supplier shall furnish and test assemblies in accordance with requirements of 4.3 (see 6.2.e).

3.4 Parts and materials. Parts and materials shall be as specified herein and as defined by applicable drawings. All material not specified shall be supplied by the contractor and shall conform to the requirements of MIL-P-11268.

3.5 Major components. The major components of the NVG shall consist of:

3.5.1 Head mounted assembly.
- (a) Objective Lens
- (b) Eyepiece Assembly
- (c) Image Intensifier Tube and Power Supply Assembly
- (d) Face Mask Assembly
- (f) Head Straps

3.5.2 Ancillary equipment.
- (a) Antifogging Devices
- (b) Carrying Case
- (c) Shipping/Storage Case
- (d) Sacrificial Filter
- (e) Protective Eyecups
- (f) Lens Cover

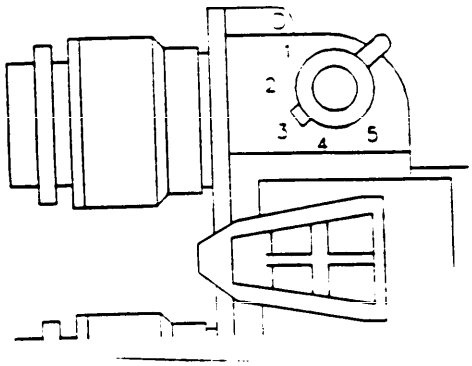
3.6 Operational and environmental.

3.6.1 Controls and adjustment.

3.6.1.1 Electrical switch. The rotary switch shall have 5 distinct positions, approximately 30 degrees between each position. When the system is removed from the face mask while in position 4 or 5 the system will automatically turn off, and will remain off until the switch is turned to position 1 or 2, or until the system is returned to the face mask while the switch is still in position 4 or 5.

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Position 1 (push button)	tube ON, IR diode ON, Indicator diode ON
Position 2	Tube ON only
Position 3	OFF
Position 4	Tube ON only
Position 5 (push button)	Tube ON, IR diode ON, Indicator diode ON



NOTE: When the goggle is attached to the face mask, battery voltage is applied to positions 4 and 5 and no battery voltage is applied to positions 1, 2, and 3. When the goggle is taken out of the face mask, battery voltage is applied to positions 1 and 2 and no battery voltage is applied to 3, 4, and 5.

3.6.1.2 Interpupillary adjustment. Each eyepiece lens assembly shall slide toward and away from the other with a force of at least 4 pounds and no greater than 10 pounds. The interpupillary distances of the eyepiece assembly shall be from 55 mm minimum to 71mm maximum.

3.6.1.3 Goggle/facemask attachment. The NVG shall have a latch mounting assembly that will permit mounting and demounting of the goggle to the face mask using one hand.

3.6.2 System performance.

3.6.2.1 Field-of-view. The system field-of-view shall be 40 degrees ± 2 degrees with a 18.0mm tube format.

3.6.2.2 Magnification. Total system magnification shall be unity (x1) ± 3%.

3.6.2.3 Linear distortion. Total system distortion shall be less than or equal to 3% across the total field-of-view. Differential distortion between channels shall be not greater than 1%.

3.6.2.4 Resolution.

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3.6.2.4.1 Far focus. The NVG shall have an on-axis resolution at the true infinity setting of not less than 0.68 cycles per milliradian and not less than 0.50 cycles per milliradian for an off-axis position of 14 degrees. The off-axis measurement shall be made at the same focus setting as the on-axis reading at optimum target illumination. The infinity focus stop shall be set to ensure that through focus adjustment is obtained. The on-axis resolution at the infinity focus stop shall not be less than 0.56 cycles per milliradian.

3.6.2.4.2 Close focus. The on-axis resolution, at a close focus of 25cm, shall not be less than 0.50 cycles per milliradian for optimum target illumination.

3.6.2.5 Collimation. The line-of-sight, as viewed through the two eyepieces, shall converge 1 degree ± 1 degree in the horizontal plane and shall be collimated within one-half degree in the vertical plane.

3.6.2.6 Focus adjustment. The NVG shall provide focus adjustment over the range from 25cm to infinity. The focus shall be continuously adjustable with one hand while wearing arctic gloves or mittens.

3.6.2.7 Brightness gain. The brightness gain shall be not less than 2000 footlamberts per footlambert. There shall be no change in brightness gain greater than 3% of the average gain when the assembly is subjected to steady input illumination.

3.6.3 Physical characteristics.

3.6.3.1 Forward projection. With the objective at the near-focus position and the eyepiece at +2 diopters, the maximum forward projection from the exit pupil shall not exceed 190mm. Minimum forward projection is desirable for all elements of the goggles.

3.6.3.2 Center of gravity. The center of gravity in a horizontal plane shall not exceed 90mm forward of the exit pupil.

3.6.3.3 Interchangeability. The objective lens assembly and the binocular eyepiece lens assembly shall be physically interchangeable between goggle assemblies without degradation of the resolution requirements (see 3.6.2.4) or the diopter focus range requirement (see 3.7.2.3). These interchanges shall be made without any modification to of any of the parts or assemblies involved.

3.6.4 Weight. Total weight of the goggle, with the BA-5567/U battery, less ancillary equipment and less coated optical component assembly, shall not exceed 680 grams.

3.6.5 Environmental conditions. The NVG and its ancillary equipment shall not be damaged and shall meet the performance requirements specified herein when subjected to the following environmental conditions:(see 3.9.1)

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3.6.5.1 High temperature. The NVG shall be capable of continuous operation up to a temperature of +45°C and storage up to a temperature of +71°C.

3.6.5.2 Low temperature. The NVG shall be capable of continuous operation or storage down to a temperature of -51°C.

3.6.5.3 Temperature shock. The NVG shall be capable of withstanding temperature shock changes occurring within 5 minutes over a temperature of 23°C to +71°C and return, and +23°C to -51°C and return.

3.6.5.4 Altitude. The NVG shall be capable of continuous operation from sea level to 15,000 feet pressure altitude and withstand a non-operating pressure altitude of 40,000 feet.

3.6.5.5 Humidity. The NVG shall operate and not be damaged when subjected to relative humidities up to 94% ± 4% and temperatures up to +65°C.(see 3.9.1)

3.6.5.6 Vibration. The assembly shall not be damaged by vibrations in the frequency range of 5 to 55Hz and up to 2.5g.(see 3.9.1)

3.6.5.7 Shock. The NVG assembly in its carrying case and in turn, in the shipping case, shall not be damaged and shall operate without degradation in performance after being dropped from a height of not less than 122cm. (see 3.9.1)

3.6.5.8 Sand and dust. The NVG shall withstand, in both operating and nonoperating conditions, exposure to fine sand and dust particles. The clear sacrificial filter shall be installed during this test to protect the objective lens.

3.6.5.9 Fungus. The NVG (except leather pads and straps) shall not support in a nonoperating condition fungus growth encountered in tropical climates and shall not be damaged by the presence of a fungus supportive environment.(see 3.9.1)

3.6.5.10 Immersion. The assembly shall not be damaged and shall operate without degradation of performance after being immersed in fresh water to a depth of not less than 3 feet for a period of not less than 30 minutes.(see 3.9.1)

3.6.5.11 Salt fog. The NVG shall withstand, in both operating and nonoperating conditions, exposure to salt-sea atmosphere and shall not be damaged.(see 3.9.1)

3.6.5.12 Explosive atmosphere. The equipment shall not cause ignition of an ambient explosive-gaseous mixture when operating in such an atmosphere.

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3.6.5.13 Transportability. The NVG shall be capable of withstanding the environment associated with wheeled or tracked vehicle transport while being carried or stowed in the carrying or shipping case.

3.6.6 Treatment and painting. Unless otherwise specified, the metallic parts of assembly shall be given a treatment and painting per MIL-STD-171.

3.6.7 Finishes. All internal surfaces except for light transmitting elements that are exposed to light from external and internal sources shall be finished to achieve the lowest feasible light reflectance. External surfaces shall be rust-resistant and self-lubricating whenever practical.

3.6.8 Electromagnetic interference. The NVG shall meet the requirements of MIL-STD-461, class A3. The required tests are RE02, wide and narrow bands, radiated emission electrical field; RS02, magnetic induction field; RS03 radiated susceptibility, electrical field.

3.7 Major component characteristics.

3.7.1 Objective lens. The objective lens shall operate with a glass faceplate that forms the input to the image intensifier tube.

3.7.1.1 f/number. The f/number of the objective lens shall be no greater than 1.20.

3.7.1.2 T/number. The T/number of the objective lens shall be no greater than 1.30. The objective lens transmission shall be designed to peak at 0.85 microns.

3.7.1.3.1 Stray light. The stray light of the objective lens shall be no greater than 2.35%.

3.7.1.3.2 Veiling glare, off axis. The objective lens when installed on the AN/PVS-7A system shall not display visible signs of ghost imagery, crescents, moons or streaking when a point light source of 2856 ± 200K is illuminating the objective lens aperture at .005 ± 0.001 fc, outside the field-of-view of the system and up to an angle of 90 degrees off the optical axis of the objective lens.

3.7.1.3.3 Veiling glare.

3.7.1.4 Equivalent focal length. The equivalent focal length of the lens assembly shall be 27.0mm ± 0.3mm.

3.7.1.5 Linear distortion. The linear distortion of the lens assembly shall be 8.0% ±.5% barrel distortion.

3.7.1.6 Field-of-view. The field-of-view of the objective shall be 40 degrees ± 2 degrees for a 18mm diameter format.

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3.7.1.7 Modulation transfer function (MTF). Over the spectral region of 630 to 900 nanometers (nm), the objective shall have MTF performance as follows:

3.7.1.7.1 MTF, axial and two-thirds field. The objective shall have an axial MTF not less than that reflected in the values below for on-axis. In the same flat focal plane chosen for axial MTF, the objective shall have two-third performance not less than reflected in Table I. If refocusing is required the deviation from a flat focal plane shall not exceed +0.00 - .005mm.

3.7.1.7.2 MTF, full field. In the same focal plane chosen for axial MTF, the objective shall have full-field performance not less than the values below for 19 degrees (full field). If refocusing is required the deviation from a flat focal plane shall not exceed +0.00 - .005mm.

TABLE I. Objective MTFs

FREQUENCY LP/MM	ON-AXIS	OFF AXIS 12.6° (2/3 FIELD) RAD.	TANG	19° (FULL FIELD) RAD.	TANG
10	94	74	74	55	62
20	84	48	39	38	28
30	65	31	21	27	16
40	61	21	15	15	10

3.7.1.8 Relative illumination. At full aperture, the image illuminance at the edge of the format shall be no less than 40% of the image illuminance on axis.

3.7.1.9 Torque. The torque required to focus the objective lens assembly shall not exceed 80in.-oz.

3.7.1.10 Weight. The weight of the lens assembly shall not exceed 72 grams.

3.7.1.11 Flange focal distance. The flange focal distance of the objective shall not be less than 1.20 millimeter measured from the rearmost flange to the input faceplate of the image intensifier assembly, when the infinity focal plane of the objective is coincident with the rear surface of the input faceplate.

3.7.2 Eyepiece lens assembly. The binocular eyepiece lens assembly shall perform as specified.

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3.7.2.1 Equivalent focal length. The equivalent focal length of the eyepiece shall be 27.0mm ± 0.3mm.

3.7.2.2 Linear distortion. The linear distortion of the eyepiece at the edge of the 18mm format shall be 8.0% ± 0.5% pin-cushion distortion.

3.7.2.3 Diopter focus. The eyepiece diopter focus shall be variable from +2 to -6 diopters, referenced to 27 ± .003mm flange focal distance (from image tube mounting surface).

3.7.2.4 Field curvature. The field curvature of the eyepiece shall fall between 0 and 1 diopter from the center of the format to an 8.75 semifield height.

3.7.2.5 Transmission. The transmission of the eyepiece shall be not less than 70% over the full eyepiece aperture for the spectral output of the P-20 phosphor, and the photopic eye (see Figure 5).

3.7.2.6 Resolution. The eyepiece assembly shall have an on-axis resolution of not less than 51 lp/mm. The 2/3 field off-axis resolution shall not be less than 36 lp/mm. The resolution measurements shall be made using a 2856K light source filtered to simulate a P-20 phosphor.

3.7.2.7 Torque. The eyepiece focus shall rotate with a torque of 70 ± 60 in.-oz. applied to the diopter adjust ring.

3.7.2.8 Exit pupil size. The exit pupil for an on-axis point in the field-of-view shall be 8 ± 0.5mm circular at a 27 ± 2mm eye relief distance.

3.7.2.9 Flange focal distance. The flange focal distance shall not be less than 4.54mm from the nearest mechanical flange.

3.7.2.10 Modulation transfer function. The binocular eyepiece shall have the following minimum axial and 2/3 field MTF for a 5mm circular stop located at the 25mm eyerelief distance.

	MODULATION TRANSFER FUNCTION		
Frequency LP/MM	AXIAL	2/3 FIELD RAD	(12.6 DEGREE) TAN
10	92	90	90
20	83	78	78
30	75	69	60
40	70	64	48

3.7.2.11 Weight. The weight of the eyepiece shall not exceed 375 grams. Includes all hardware except batteries.

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3.7.2.12 Collimation. The conjugate beam emerging from the two eyepieces shall diverge at one degree ± one degree in the horizontal plane, and be collimated within 1/2 degree in the vertical plane.

3.7.3 Battery. The NVG shall be powered by the following: two BA-3058/U batteries conforming to MIL-B-49030/6, or one BA-5567/U battery conforming to MIL-B-49430/4. The system shall not be damaged by incorrect insertion of any of the specified batteries.

3.7.4 Light emitting diode (LED). The NVG shall be equipped with a forward projecting LED infrared source (wavelength of LED shall be 880 ± 20nm).

3.7.5 LED indicator. A visual light shall be presented in the left eyepiece to indicate to the operator that the LED is energized.

3.7.6 Face mask and cushion. The face mask shall be compatible with the M1, CVC, and DH-145 helmets and the M17A1, M17A2, M24, M40, M25A1, XM30, XM34, XM40, XM41, XM42 and XM43 protective masks. The face mask shall provide a quick disconnect and both a horizontal and vertical adjustment for the NVG.

3.7.7 Demist shields. Demist shields shall be provided for each eyepiece. The demist shields shall clip-on when needed and be removed easily when not needed.

3.7.8 Sacrificial filter. A sacrificial filter shall be provided for the objective lens assembly for protection against sand and dust. The filter shall provide at least 90% transmittance and, with the objective meet the stray light requirement of the objective.

3.7.9 Eye cups. Eye cups shall be provided for each eyepiece.

3.7.10 Shipping and storage case. A shipping and storage case shall be provided. The case shall provide protection to ensure that the NVG will meet the shock and transportability requirements of paragraphs 3.6.5.7 and 3.6.5.13.

3.7.11 Carrying case. A lightweight carrying case shall be provided and shall provide protection in a field environment.

3.8 Workmanship. The NVG assembly shall conform to Requirement 9, MIL-STD-454. Joints and seams shall be tight fit, and electrical wiring shall be secure and with unbroken insulation. Padding shall be free of gouges, tears, excess adhesive, or imbedded foreign matter and shall be attached firmly to the proper surface. All objective lens and eyepiece assemblies shall be free from cracks, splits, cold flow, shrinkage, inclusions, porosity, or any similar characteristics. Threads shall be full and undamaged for the entire length or depth as required on the applicable drawing. Objective lens and eyepiece assemblies shall be free from burrs and shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

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The cleaning methods used shall not damage any of the objective lens or eyepiece assemblies nor shall the parts be contaminated by the cleaning agent. Optical quality requirements defined as scratches, digs, edge chips, bubbles, coating defects and cementing defects shall be in accordance with MIL-O-13830 except that inspection of optical cleanliness shall be with the unaided eye (without 3X magnification). All moving parts and adjustments shall be examined to insure that they move freely throughout their entire range without sticking, binding, or creeping.

3.9 Technical interpretations. The following technical interpretations are, when referenced in sections 3, 4, or 5, mandatory for this specification.

3.9.1 Damage. Damage is defined as an operational failure or malfunction including arcing, corona, intermittent operation, flickering, brittleness, cracking, corrosion, or failure of any finish; condensation, fogging, or moisture on any internal surface; dents, or breakage of any hardware, connection or component; or any change in collimation exceeding requirements.

3.9.2 Ambient temperature. Ambient temperature is defined as +23°C, +10°C, -2°C.

3.9.3 Qualified product. A product which has successfully met all of the requirements of and IPT or first article witnessed by the procuring activity.

3.9.4 Environmental tests. For the purpose of this specification, environmental tests are those tests in paragraph 4.6.4.

3.9.5 Refurbished . Correcting all damage to the item or any part of the item. Restoring the item to a fully operational and acceptable status.

3.10 Nameplates and product marking. Items of material provided by the contractor shall be identified in accordance with MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and test. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection and test requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection and test requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections and/or tests set forth in the specification where such inspections and/or tests are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections and tests. Inspections and tests shall be classified as follows:

a. Initial production test (4.3)
b. Quality conformance inspection (4.4)

4.3 Initial production tests.

4.3.1 Inspection. Unless otherwise specified in the contract or purchase order, ten initial production test units shall be inspected in accordance with Table II. Presence of one or more defects shall be cause for rejection of that unit and may be cause for rejection of the initial production test.

TABLE II. Inspection.

Inspection	Requirement Paragraph	Method
Parts, material and components missing or not as specified	3.2, 3.4	visual/gages
Major component list	3.5	visual
Treatment and painting	3.6.6	visual
Finishes	3.6.7	visual
Nameplates and product marking	3.10	visual
Workmanship	3.8	visual

4.3.2 Initial production tests. Following successful completion of the inspections specified in 4.3.1, initial production test assemblies with intensifiers which have passed required tests shall be subjected to all inspections in Table III. Inspections listed in Table III may be performed in any order except that system resolution shall be performed last. Failure of any inspection shall be cause for rejection of that assembly and may be cause for rejection of the initial production test.

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TABLE III. Initial production tests.

Inspection	Requirement Paragraph	Test Paragraph
Objective Lens:		
f/number	3.7.1.1	4.6.6.1
T/number	3.7.1.2	4.6.6.2
Veiling glare	3.7.1.3	4.6.6.3
Equivalent focal length	3.7.1.4	4.6.6.4
Linear distortion	3.7.1.5	4.6.6.5
Field of view	3.7.1.6	4.6.6.6
Modulation transfer function	3.7.1.7	4.6.6.7
Relative illumination	3.7.1.8	4.6.6.8
Weight	3.7.1.10	4.6.6.9
Eyepiece Lens:		
Equivalent focal length	3.7.2.1	4.6.7.1
Linear distortion	3.7.2.2	4.6.7.2
Field curvature	3.7.2.4	4.6.7.3
Transmission	3.7.2.5	4.6.7.4
Resolution	3.7.2.6	4.6.7.5
Exit pupil	3.7.2.8	4.6.7.5
System Performance:		
Electrical switch	3.6.1.1	4.6.1.1
Interpupillary adjustment	3.6.1.2	4.6.1.2
Field of view	3.6.2.1	4.6.2.1
Magnification	3.6.2.2	4.6.2.2
Linear distortion	3.6.2.3	4.6.2.3
Resolution	3.6.2.4	4.6.2.4
Collimation	3.6.2.5	4.6.2.5
Objective focus (torque)	3.7.1.9	4.6.2.6
Diopter focus and (torque)	3.7.2.3, 3.7.2.7	4.6.2.7
Brightness gain	3.6.2.7	4.6.2.8
Weight	3.6.4	4.6.3.1
Forward projection	3.6.3.1	4.6.3.2
Center of gravity	3.6.3.2	4.6.3.3
Diopter focus (range)	3.7.2.3	4.6.2.7
Objective focus (range)	3.6.2.6	4.6.2.4
LED	3.7.4	4.6.2.9
LED indicator	3.7.5	4.6.2.10
Interchangeability	3.6.3.3	4.6.3.4
Environmental:		
High temperature	3.6.5.1	4.6.4.1
Low temperature	3.6.5.2	4.6.4.1
Temperature shock	3.6.5.3	4.6.4.2
Altitude	3.6.5.4	4.6.4.1
Humidity	3.6.5.5	4.6.4.3
Vibration	3.6.5.6	4.6.4.4
Shock	3.6.5.7	4.6.4.5
Sand and dust	3.6.5.8	4.6.4.6

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TABLE III. Initial production tests - (Continued)

Inspection	Requirement paragraph	Test paragraph
Fungus	3.6.5.9	4.6.4.7
Immersion	3.6.5.10	4.6.4.8
Salt fog	3.6.5.11	4.6.4.9
Explosive condition	3.6.5.12	4.6.4.10
Transportability	3.6.5.13	4.6.4.11
EMI	3.6.8	4.6.5

4.4 Quality conformance inspections and tests.

4.4.1 Inspections. Materials, parts, processes, and assemblies shall be examined for conformance to the applicable specification or drawing and for the requirements of Table IV. Failure of an assembly to pass these inspections shall be cause for rejection of that assembly.

TABLE IV. Quality conformance inspection.

Inspection	Method
Workmanship	visual
Nameplates and product marking	visual
Parts, materials and components missing or not as specified	visual/gages
Treatment and painting	visual

4.4.2 Lens inspection. The objective and eyepiece sub-assemblies shall be tested in accordance with Table V. For purposes of Group A and Group B testing, a lot shall be determined by the contractor. For Group A testing the entire lot shall be inspected. Group B testing the lot shall be sampled as directed by the contract or purchase order.

TABLE V. Lens inspection.

Test	Requirement Paragraph	Test Paragraph
Objective Lens Assembly		
Group A Test		
MTF	3.7.1.7.1	4.6.6.7
Group B Subgroup 1		
EFL	3.7.1.4	4.6.6.4
f/No.	3.7.1.1	4.6.6.1
T/No.	3.7.1.2	4.6.6.2
Veiling glare	3.7.1.3	4.6.6.3

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TABLE V. Lens inspection - (Continued)

Test	Requirement paragraph	Test paragraph
Weight	3.7.1.10	4.6.6.9
MTF	3.7.1.7.2	4.6.6.7
Group B Subgroup 2		
Field of view	3.7.1.6	4.6.6.6
Relative illum.	3.7.1.8	4.6.6.8
Linear distortion	3.7.1.5	4.6.6.5
Eyepiece Lens Assembly		
Group A Test		
Resolution	3.7.2.6	4.6.7.5
Group B Subgroup 1		
EFL	3.7.2.1	4.6.7.1
Transmission	3.7.2.5	4.6.7.4
Diopter focus	3.7.2.3	4.6.2.7
Group B Subgroup 2		
Linear distortion	3.7.2.2	4.6.7.2
Field curvature	3.7.2.4	4.6.7.3
Exit pupil	3.7.2.8	4.6.7.5

4.4.3 NVG Group A inspection. Each NVG shall be tested in accordance with Table VI. Failure of an NVG to pass these tests shall be cause for rejection of that system.

TABLE VI. Group A inspections.

Test	Requirement Paragraph	Test Paragraph
Electrical/switch	3.6.1.1	4.6.1.1
Resolution	3.6.2.4	4.6.2.4
Objective focus (range)	3.6.2.6	4.6.2.4
LED	3.7.4	4.6.2.9
LED indicator	3.7.5	4.6.2.10
Collimation	3.6.2.5	4.6.2.5

4.4.4 Group B inspection. Systems which have passed the inspection of Group A shall be formed into lots. From these lots, sample assembles shall be selected and subjected to the inspections of Table VII. Sampling shall as per the contract or purchase order.

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TABLE VII. Group B inspections.

Test	Requirement Paragraph	Test Paragraph
Interpupillary adjustment	3.6.1.2	4.6.1.2
Brightness gain	3.6.2.7	4.6.2.8
Objective focus (torque)	3.7.1.9	4.6.2.6
Diopter focus (torque	3.7.2.7	4.6.2.7

4.4.5 Group C inspections. Group C inspections shall be performed on assemblies that have passed Group A, or Group A and B inspections. Samples shall be as specified in the contract or purchase order.

TABLE VIII. Group C inspections.

Inspection	Requirement paragraph	Test paragraph
Subgroup I		
High temperature	3.6.5.1	4.6.4.1
Low temperature	3.6.5.2	4.6.4.1
Temperature shock	3.6.5.3	4.6.4.2
Altitude	3.6.5.4	4.6.4.1
Immersion	3.6.5.10	4.6.4.8
Subgroup II		
Vibration	3.6.5.6	4.6.4.4
Shock	3.6.5.7	4.6.4.5
Transportability	3.6.5.13	4.6.4.11
Subgroup III		
Weight	3.6.4	
Forward projection	3.6.3.1	4.6.3.1
Center of gravity	3.6.3.2	4.6.3.2
Interchangeability	3.6.3.3	4.6.3.3
Distortion	3.6.2.3	4.6.2.3
Subgroup IV		
Humidity	3.6.5.5	4.6.4.3

4.4.5.1 Group C failures. Actions required in the event of a Group C failure shall be as specified in the contract or purchase order (see 6.2.c).

4.4.5.2 Disposition of Group C inspection units. Assemblies that have been subjected to the inspections listed in Table VIII shall be refurbished (see 3.9.5) as necessary and subjected to the controls and adjustments inspection of paragraph 4.6.1 as a minimum.

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If refurbishment requires disassembly, the resolution, diopter focus and focus adjustment inspections shall be performed. Failure to pass these requirements shall be cause for rejection of that assembly.

4.5 Test conditions.

4.5.1 Test methods. Tests shall be conducted in accordance with the test procedures specified herein. Unless otherwise specified, the following conditions shall apply:

a. The radiation source used in the tests shall be a tungsten filament lamp operated at a color temperature of 2856K ± 50K.

b. The photometer used for screen brightness measurements shall be a Pritchard Model 1980 PR, or equal.

c. The photometer used for brightness measurements shall be calibrated against a standard source as specified below:

(1) Tungsten filament lamp operated in conjunction with an opal glass such that the color temperature of the radiation emitted from the opal glass is 2856K ± 50K.

(2) Corning spectral filters Nos. 3-71 and 4-67, or equivalent.

(3) Opal glass to produce a uniform, Lambertian distribution.

(4) Output brightness to be 0.1 to 1.0 footlambert uniformly distributed.

d. Tolerance on specified radiation levels shall be ± 10%.

e. Meters used for monitoring lamp current and voltage shall have an accuracy of ± 0.25%.

f. Neutral density filters used in test equipment shall have transmission characteristics within 10% of the nominal filter transmission from 0.35 to 1.0 micrometer.

g. Tests shall be performed at ambient conditions as defined in 3.9.2 except where otherwise specified.

h. Test chambers used for environmental temperature tests shall maintain the temperature with ± 2°C.

i. Resolution targets shall be a positive, nominal 100% contrast USAF 1951 resolving power test target.

j. Alternate test methods that are found acceptable by the Government may be used.

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4.6 Operational and environmental. All operational and environmental tests (see 3.9.4) requiring performance check may have the check conducted with batteries BA-5567/U or BA-3058/U. A power supply with an output of 2.7 ± 0.05 Vdc may be used for testing purposes. Each NVG assembly shall demonstrate its ability to perform properly after being subjected to all the environmental tests. For the purpose of post environmental testing, "perform properly" shall be defined as proper operation of controls and adjustments (3.6.1.1, 3.6.1.2, 3.7.1.10, and 3.7.2.7) and the ability to successfully compete the resolution (3.6.2.4) test.

4.6.1 Controls and adjustments.

4.6.1.1 Electrical/switch. The NVG shall be mounted in the head mount and examined for compliance with the requirements of 3.6.1.1. Remove the NVG from the head mount and examine for compliance with the requirements of 3.6.1.1. Failure of the assemblies to meet the requirements of 3.6.1.1 shall constitute failure of the test.

4.6.1.2 Interpupillary adjustment. Using a force gage or other suitable method measure the force required to slide each eyepiece toward or away from each other and record. Set the eyepiece IPD adjustment to its minimum. Using a caliper, measure the distance between the outer edge of the left diopter focus ring and the outer edge of the right diopter focus ring. Subtract the diameter of either diopter focus ring and record it. Set the eyepiece IPD to its maximum. Repeat the measurement and record it. Failure to meet the force and IPD requirements of 3.6.1.2 shall constitute failure of this test.

4.6.2 System performance.

4.6.2.1 Field of view. The field of view of the NVG shall be measured using a rotary turntable and a vertical line that is either projected through a collimator or wall mounted. With the NVG mounted to the rotary turntable, rotate the NVG so that the vertical line can just be seen at either side of the FOV. Record the reading on the rotary table vernier scale then rotate the NVG so that the vertical line can just be seen at the opposite FOV limit. Record the reading on the vernier scale. The difference in the two readings is the FOV. Failure to meet the requirements of 3.6.2.1 shall constitute failure of this test.

4.6.2.2 Magnification. Magnification of the goggle shall be measured with the compound rotary table method.

(1) Both rotary tables are set to their zero positions and the image of the collimator target is super imposed on the cross hairs in the sighting telescope.

(2) A small angular rotation of Table A is made of A degrees, where A = 1/10 x minimum FOV/2.

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(3) Table B, attached to table A in then rotated B degrees until the image of the collimator target is returned to the sighting telescope cross hairs. Angle B is then recorded.

(4) Take a similar reading at 2A.

(5) Readings are then taken for negative angles -A, -2A.

(6) The goggle system magnification, M(A), at each angular displacement is then calculated by:

$$M(A) = \tan (B)/\tan (A)$$

(7) The mean of four values of M(A) taken at $\pm$ -A, and $\pm$ -2A is calculated an recorded as the system magnification. Failure to meet the requirements of 3.6.2.2 will constitute failure of this test.

4.6.2.3 Linear distortion. The linear distortion of the NVG shall be measured using a collimator to project a crosshair target, dioptometer, and a turntable with vernier scale graduated in degrees/minutes. Mount the NVG to the turntable by means of a suitable holding fixture that will permit positioning and supporting the dioptometer behind each of the two eyepieces. The dioptometer must be capable of being rotated independently of the NVG. Energize NVG and with dioptometer, set each eyepiece to 0. Set NVG for best focus of projected target. Rotate the NVG to the edge of the left side of field of view (approximately 19 degrees) while viewing through dioptometer. Superimpose dioptometer crosshair over projected crosshair. Record reading off of holding fixture rotary table scale as R1. Set test bench rotary table vernier scale to 0. Rotate table to 38 degrees to right side of field of view. Swing dioptometer until dioptometer crosshair is again superimposed on projected crosshair. Record reading off of dioptometer rotary table scale as R2.

Compute the system distortion as follows:

$$\text{Distortion} = \frac{\text{Tan } \theta}{\text{M Tan } 19^{o}} - 1 \times 100$$

Where:
M = System magnification

$\theta = \frac{R2-R1}{2}$ = Average of right and left of diopter scope for 38° of system rotation (system field of view).

Center the remaining eyepiece assembly to the crosshair target and repeat test as described above.

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Failure to meet the requirement of 3.6.2.3 shall constitute failure of this test.

4.6.2.4 Resolution. Place a 100% contrast USAF 1951 Resolving Power Test Target (black on a clear background) in the infinity focal plane of a collimator having a known focal length, back illuminate the target with 2856K white light of such intensity to achieve maximum system resolution. Position NVG in holder on rotary table with objective lens looking into collimator. Adjust NVG for best focus. Read resolution on axis and record reading. Adjust rotary table 14 degrees off-axis and without refocussing, read resolution and record reading. Set the resolution target at 25cm from the assembly and determine the smallest pattern that can be resolved. Failure to meet the requirements of 3.6.2.4 shall constitute failure of this test.

4.6.2.5 Collimation. Collimation of the NVG shall be measured either by the method described below or an approved equivalent. The test equipment shall consist of a test set as illustrated in Figure 1, and a target reticle as illustrated in Figure 2. A fixture, suitable for supporting the NVG shall be mounted such that the NVG is approximately centered on the optical axis of the collimator. A collimation fixture consisting of two parallel telemicroscopes is mounted to a turntable immediately behind and aligned to each eyepiece of the NVG. Prior to installing the NVG, observe that the crosshair of right side telescope is aligned to the projected "crosshair". Adjust telescope as required to achieve alignment. Check to insure that the fixed crosshair reticle in the filar eyepiece of the left telescope is also aligned to the projected crosshair. Prior to installing the NVG on the holding fixture, adjust objective for infinity focus by focusing on the projected collimator target and adjust both eyepieces to"0" diopters using a diopter scope or adjust both eyepieces for best focus while looking through collimation fixture. Attach the NVG to the holding fixture and adjust the interpupillary distance to match the two telescopes. Attach viewer to the holding fixture. Energize the viewer. View through the right and left telescope individually. Position the fixed crosshair of the right and left telescope, using the azimuth and elevation adjustment of the test bench, so the crosshairs are located inside the tolerance zone of the projected target. The maximum limits for divergence and divergence are represented by the solid lines on either side of the solid "crosshair" (see Figure 2). Failure of the intersection point of the crosshairs in the right and left telescopes to fall within the vertical and horizontal tolerance limits of the projected target constitutes failure to meet the requirements 3.6.2.5 and constitutes failure of this test.

4.6.2.6 Objective focus. The assembly focus range adjustments on the objective lens shall be rotated not less than 3 times throughout the total range of movement, and returned to the close focus (25 cm) position stop. A suitable torque measuring device shall be used to determine the breakaway torque of the focus range adjustment. Failure to meet the requirements of 3.7.1.10 shall constitute failure of the test.

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4.6.2.7 Diopter focus. Measure the diopter adjustment of each eyepiece at each extreme of travel using a diopterscope whose entrance aperture is set at the eye relief distance. Rotate each eyepiece through its full adjustment range of not less than 3 times. Measure the torque of each monocular via the diopter adjustment ring over full range of travel in each direction. Failure to meet the requirement of 3.7.2.3 and 3.7.2.7 shall constitute failure of this test.

4.6.2.8 Brightness gain. Place a 2856K extended lambertian light source at least 25cm in front of the assembly objective. This extended lambertian source must be large enough to represent a 40 degree field-of-view to the assembly (i.e., the entire photocathode format is illuminated). Using a calibrated photometer, the extended light source shall be set to a brightness of not more than $5X10^{-4}$ footlamberts. Activate the assembly looking at the extended source and read output through the eyepiece using a photometer calibrated to read correctly with a 5mm entrance pupil. This 5mm entrance pupil shall be placed on the optical axis of the photometer and the optical axis of the assembly at the specified at the system exit pupil. Gain is calculated as:

$$\text{Gain} = \frac{\text{Output brightness}}{\text{Input brightness}}$$

Failure to meet the requirement of 3.6.2.7 shall constitute failure of this test.

4.6.2.9 LED. The LED operation shall be tested by verifying operation through visual indication with the LED energized in a darkened room. Failure to meet the requirements of 3.7.4 shall constitute failure of this test.

4.6.2.10 LED indicator. While performing the LED operation test, 4.6.2.9, observe that the LED indicator light in the left eyepiece is illuminated. Failure of the indicator light to be illuminated in accordance with 3.7.5 when the LED light source is energized shall constitute failure of this test.

4.6.3 Physical characteristics. The following characteristics shall be measured with the NVG, including the facemask, and with the BA-5567/U battery installed.

4.6.3.1 Weight. The NVG, less ancillary equipment, and less coated optical component but with the BA-5567/U battery installed, shall be weighed on a scale accurate to 1 gram. Failure to meet the requirement of 3.6.4 shall constitute failure of this test.

4.6.3.2 Forward projection. With the optical assembly installed in the foremost projection position and the objective lens at the near focus position, and the eyepiece at + 2 diopters, the distance from the exit pupils to the most forward project shall be measured with a measuring instrument to an accuracy of 0.1mm.

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Failure to meet the requirements of 3.6.3.1 shall constitute failure of this test.

4.6.3.3 Center of gravity. The center of gravity (C.G.) of the NVG, including headmount assembly and BA-5567/U battery installed, shall be measured with the objective focus set at near focus and the eyepiece set at full negative diopter setting. Determine the C.G. of the NVG by balancing the NVG on a fulcrum. Measure the distance from the fulcrum to the 15mm exit pupil of the eyepiece. Failure to meet the requirement of 3.6.3.2 shall constitute failure of this test.

4.6.3.4 Interchangeability. All IPT assemblies which have passed the resolution requirement (see 3.6.2.4) and the diopter focus range requirement (see 3.7.2.3) shall be tested for interchangeability. For this test, the mount assemblies, objective lens assemblies, and binocular eyepiece lens assemblies shall be interchanged between goggle assemblies as shown in Figure 6 and the resolution (see 4.6.2.4) and diopter focus range (see 4.6.2.7) tests repeated after each test arrangement and after reassembly to the pretest arrangement. This test shall be repeated for all additional IPT assemblies until each IPT assembly has been tested once as shown in Figure 6 with another IPT assembly. Failure to meet the requirements of 3.6.3.4, 3.6.2.4, and 3.7.2.3 after each arrangement shall constitute failure of this test.

4.6.4 Environmental tests. Environmental tests performed on the NVG shall follow the general procedures of MIL-STD-810 for ground based equipments. Failure to pass any environmental requirement shall constitute test failure.

Prior to and following each environmental test and at such times during environmentals when the procedure calls for a "test," the following tests shall be performed:

a. The NVG shall operate.

b. All controls and adjustments shall operate smoothly.

c. High light level resolution test (4.6.2.4).

4.6.4.1 Temperature/altitude. The NVG shall be tested in accordance with Method 518.1 of MIL-STD-810 with the following variations:

(1) Humidity shall be performed as a separate test.

(2) The low temperature extreme shall be -51°C.

(3) The maximum high temperature shall be as follows:

Operating	45°C
Non-operating	71°C

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(4) Maximum operating altitude shall be 15,000 feet.

(5) Maximum non-operating altitude shall be 40,000 feet.

Failure to meet the requirements of 3.6.5.1, 3.6.5.2, and 3.6.5.4 shall constitute failure of this test.

4.6.4.2 Temperature shock. The NVG shall be subjected to Procedure I, Method 503.1 of MIL-STD-810 for compliance with 3.6.5.3 except the shock temperature shall be from +23°C to +71°C and from +23°C to -51°C. Failure to meet the requirement of 3.6.5.3 shall constitute failure of this test.

4.6.4.3 Humidity. The NVG shall be subjected to Procedure II, Method 507.1, Figure 507.1-2 of MIL-STD-810 except as follows:

(a) Omit steps 1, 2, 3, 7 and 9.

(b) Step 5 shall be as-is.

(c) Step 6 as-is, except for performance of inspection for operation shall be during the last four hours of the third cycle only.

(d) Step 8 inspect for operation and inspection for damage or leakage shall be made not less than 4 hours after the end of the last cycle.

Operational tests shall include controls and adjustments and resolution. Failure to meet the requirement of 3.6.5.5 shall constitute failure of this test.

4.6.4.4 Vibration. The NVG shall be subjected to Procedure X, Method 514.2 of MIL-STD-810. The NVG removed from its carrying case and shipping case shall be subjected to vibration profile AV of Table 514.2-VII and Figure 514.2-7. Failure to meet the requirements of 3.6.5.6 shall constitute failure of this test.

4.6.4.5 Shock. The assembly in its carrying case and in turn, in its shipping/storage case shall be subjected to Procedure II, Method 516.2 of MIL-STD-810. Failure to meet he requirements of 3.6.5.7 shall constitute failure of this test. Shipping case dents or finish damage shall not constitute failure providing the protective function of the case is not impaired. Cases with such damage may be re-used for shock (drop) testing.

4.6.4.6 Sand and dust. The NVG shall be subjected to Procedure I, Method 510.1 of MIL-STD-810 for compliance with the requirements of 3.6.5.8. This test shall be run with sacrificial filter installed per 3.7.8. Failure to meet the requirements of 3.6.5.8 shall constitute failure of this test.

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4.6.4.7 Fungus. The NVG shall be subjected to Procedure I, Method 508.1 of MIL-STD-810 for compliance with the requirements of 3.6.5.9. Failure to meet the requirements of 3.6.5.9 shall constitute failure of this test.

4.6.4.8 Immersion. The assembly less the head mount and head straps shall be subjected to Test Method 512.1, Procedure I of MIL-STD-810 except that the immersion time shall be 30 minutes minimum. Failure to meet the requirements of 3.6.5.10 shall constitute failure of this test.

4.6.4.9 Salt fog. The NVG shall be subjected to Procedure I, Method 509.1 of MIL-STD-810 for compliance with 3.6.5.11 the lens may be protected during this test. Failure to meet the requirements of 3.6.5.11 shall constitute failure of this test.

4.6.4.10 Explosive conditions. The NVG shall be subjected to Procedure I, Method 511.1 of MIL-STD-810 for compliance with 3.6.5.12. Failure to meet the requirements of 3.6.5.12 shall constitute failure of this test.

4.6.4.11 Transportability. The NVG, in its carrying and shipping cases, shall be subjected to Procedure XI (Part 2), Method 514.2 of MIL-STD-810 for compliance with 3.6.5.13. Dents and finish damage to the shipping case and inserts shall not constitute failure, providing the use and protective function of the case is not impaired.

4.6.5 Electromagnetic radiation. The NVG shall be tested in accordance with MIL-STD-462 for compliance with the emission and susceptibility requirements of 3.11. In RS03 the contractor shall determine the threshold level of susceptability for each frequency range indicated in a thru d. The level of susceptibility shall be equal to or greater than the levels exhibited by the AN/PVS-7A, delivered under contract DAAB07-85-C-E033 using K and M power supplies.

The frequency to be tested under RE02 shall be 14KHz - 10GHz. The limitations on RE02 are as follows:

20.5 - 35.5 KHz - 35dB
35.5 - 88.0 KHz - 45dB
88.0 - 130.0 KHz - 25dB
130.0 -340 KHz - 20dB

The Radiated Susceptibility frequencies under RS03 are as follows:

a. 10KHz - 2MHz
b. 2MHz - 30MHz
c. 30KHz - 10GHz
d. 10GHz - 40GHz

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4.6.6 Lens assembly, objective. The objective shall be tested through an input glass faceplate. The glass plate shall be 5.537, ± 0.050 mm thick Corning 7065 glass, or equivalent. The two surfaces of the glass plate shall be flat to with 1/4 wavelength at 632.8 nanometers, and parallel within 2 arc-seconds. The following performance parameters, unless otherwise specified, shall be met when the image lies on the rear surface of this glass plate. Unless otherwise specified, the following requirements will be met at ambient temperature (see 3.9.2).

4.6.6.1 f/number. The f/number of the objective lens shall be calculated by dividing the equivalent focal length by the effective aperture. The effective aperture is measured with a low power microscope viewing the front of the lens aperture an determining the size of the aperture stop located inside the lens under test. Failure to meet the requirements of 3.7.14 shall constitute failure of this test.

4.6.6.2 T/number. The T/number is the equivalent f-number of a fictitious lens having a 100% transmission that produces the same central photocathode illumination, as the lens under test. The T-number is measured by placing the lens between a collimated light source and integrating sphere incorporating a detector. The signal level is measured. The signal level must be 100:1 above background, and the light source/detector combination must be in accordance with figure 3. An iris diaphragm shall be place one focal length from the entrance of the sphere, and its diameter adjusted until the signal level is the same as without the lens. The T-number is then, the effective focal length, divided by the diameter of the iris diaphragm. Failure to meet the requirements of 3.7.12 shall constitute failure of this test.

4.6.6.3 Veiling glare.

4.6.6.3.1 Stray light. Stray light glare is the percent energy that from a one hundred eighty (180) degree white surrounding field enters a one degree black spot. The stray light shall be measured with an integrating sphere where the lens under test is placed at the entrance of the sphere on the sphere's equator and at 180 degrees from the entrance a black spot is located. A radiometer consisting of a small pinhole (0.05mm) shall be used over a detector apparatus. The light source/detector combination shall be in accordance with figure 3. The light source shall be a tungsten filament lamp. The radiance of the surrounding sphere shall be greater than one hundred times the black spot. The aerial image of the sphere with the black spot is scanned for minimum signal level (i.e. finding the black spot). The geometry of the scanning pinhole relative to the detector position shall not be less than 120 degrees. A 100% level is set by replacing the black spot with a white surface. With the white spot removed the radiometric reading is the veiling glare. The light source/detector combination shall be in accordance with figure 3. Failure to meet the requirements of 3.7.1.3 shall constitute failure of this test.

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4.6.6.3.2 Veiling glare, off axis. The objective lens shall be installed on an AN/PVS-7A and a point source of color temperature 2856 ± 200K shall be placed at the edge of the field-of-view of the AN/PVS-7A. The light source shall illuminate the lens entrance pupil at a level of 0.005 ± 0.001 fc. The light source shall be placed at least 10 feet away. The brightness of the surrounding field shall be 1.0 ± 0.5 E-4 fc. Failure to meet the requirements of 3.7.1.3 shall constitute failure of this test.

4.6.6.4 Equivalent focal length. The objective lens with the simulated photocathode substrate is placed on a goniometric table that includes a measuring microscope. The light source is a collimated beam, with a pinhole or slit target, of sufficient bundle size as to fill the lens under test at all possible field angles. The size of the pinhole/slit width as geometrically projected through the lens under test shall not be greater than 0.005mm. The objective lens/photocathode substrate is aligned to the plane of the microscope measuring ways. The microscope measuring ways are previously aligned to the point source or slit source target. The Equivalent Focal Length (EFL) of the objective lens assembly is determined by measuring the image translation produced by rotating the goniometer though an angle of ± one (1) degree. the Equivalent focal length is the quotient of the difference in the measuring microscope readings divided by two over the tangent of one degree. Failure to meet the requirements of 3.7.1.4 shall constitute failure of this test.

4.6.6.4.1 EFL alternate. The equivalent focal length of the objective lens shall be measured using the system magnification method. A known target size will be projected through a collimator of a known focal length. The target size will be measured at the image plane of the objective and the EFL calculated using the known values plus the image size as measured at the image plane of the objective. The EFL will be calculated with the following equation: OBJECTIVE EFL = CI/T

Where: C - Collimator EFL
I - Measured image size
T - Projected target size

Failure to comply with the requirements of 3.7.1.4 shall constitute failure of this test.

4.6.6.5 Linear distortion. Using the set-up as described for the effective focal length test, measure the semi-field angle for a semi-field height of plus and minus nine millimeters. Using the average of the semi-field angle (sfa), the percent distortion is 100*(9.0-efl*tan(sfa)) / (efl*tan(sfa)). Failure to meet the requirements of paragraph 3.7.1.5 shall constitute failure of this test.

4.6.6.6 Field of view (FOV). The field of view of the objective shall be measured as part of the linear distortion test, 4.6.3.5. Failure to meet the requirements of 3.7.1.6 shall constitute failure of this test.

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4.6.6.7 Modulation transfer function (M.T.F.). The objective shall be tested on an MTF apparatus that yields sine wave MTF response. The MTF apparatus shall have a light source and detector whose combination shall yield a spectral response curve complying with figure 3. (If a microscope objective is used to relay the image of the objective lens, the numerical aperture (NA) of the microscope objective shall nor be less than 0.65). The MTF apparatus shall allow the objective to be tested either continuously or at a minimum of 4 distinct. Evenly distributed points from 0 to 40 lp/mm. The area analyzed shall be at least 0.5 mm in diameter. The calibration of the MTF apparatus shall be performed before the test date is valid. The calibration procedure to be used is to measure the MTF of a known size target (e.g. slit or pinhole). All references to size are at second surface of the simulated photocathode. The MTF of a pinhole is known mathematically as a first order Bessel Function. MTF = $(2J_1(x))/x$ where x= 3.14156*(lp/mm)*(pinhole diameter in mm). The MTF of a pinhole goes to zero at a spatial frequency 1.22 times its reciprocal diameter. The MTF of a slit is known mathematically as (sine (x))/(x), where x=3.14156*(lp/mm)*(slit width). The units for x is in radians. The MTF of a slit goes to zero at a spatial frequency at its reciprocal width. The procedure shall be to measure the MTF of the lens and then measure the MTF of the lens with a very large pinhole/slit.
In the case of microprocessor based MTF apparatus which correct for the pinhole/slit size, the size of the pinhole/slit entered into the computer software, when measuring the lens with large pinhole/slit should not be greater than 1/40 of the larger pinhole/slit. The geometrical size of the image shall be 0.030mm ± 0.001mm, at the second surface of the simulated photocathode. The size of the slit shall be 0.025 ± 0.001mm, at the second surface of the simulated photocathode. The MTF of the large pinhole/slit can be determined by taking the ratio of the MTF curves of the lens with large pinhole/slit, divided by the MTF of the lens under test. Differences in Modulation of greater than three percent will invalidate the MTF results. See Figure 4 for MTF as a function of spatial frequency and pinhole diameter/slit width. Failure to meet the requirements of 3.7.1.7 will constitute failure of this test.

4.6.6.8 Relative illumination. The relative illumination shall be measured with a set-up that consists of a collimated light source, goniometer, and an integrating sphere incorporating a detector. The light source/detector combination shall be in accordance with figure 3. The relative signal ratio as a function of angle to the axial signal, is the relative illumination. Failure to meet the requirements of 3.7.1.8 shall constitute failure of this test.

4.6.6.9 Weight. The objective lens assembly shall be weighted on a scale accurate to 1 gram. Failure to meet the requirements of 3.7.1.10 shall constitute failure of this test.

4.6.6.10 Flange focal distance. The flange focal distance of the objective shall be measured as part of the on axis MTF test, 4.6.2.5. Failure to meet the requirements of 3.7.1.11 shall constitute failure of this test.

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4.6.7 Lens assembly, binocular eyepiece. All optical tests on the binocular eyepiece assembly will be run on individual binocular eyepiece assemblies set at zero diopters for an object in the nominal focal plane. The light source/detector combination shall be in accordance with figure (5).

4.6.7.1 Equivalent focal length. The binocular eyepiece lens assembly with a simulated eye pupil of five (5) mm diameter at the eyerelief distance, shall be placed on a goniometric table that includes a measuring microscope. The light source is a collimated beam, with a pinhole or slit target, of sufficient bundle size as to fill the lens under test at all possible field angles. The size of the pinhole/slit width as geometrically projected through the lens under test shall not be greater than 0.005mm. The microscope measuring ways are previously aligned to the point source or slit source target. The equivalent focal length (EFL) of the binocular eyepiece lens assembly is determined by measuring the image translation produced by rotating the goniometer though an angle of ± one (1) degree. The equivalent focal length is the quotient of the difference in the measuring microscope readings divided by two over the tangent of one degree. Failure to meet the requirements of 3.7.2.1.4.1 shall constitute failure of this test.

4.6.7.2 Linear distortion. Using the set-up as described for the effective focal length test, measure the semi-field angle for a semi-field height of plus and minus 9.0 millimeters. In order to simulate the tube conditions move the microscope 3.23 ± .10mm toward the eyepiece. Using the average of the semi-field angle (sfa), the percent distortion is 100*(9.00-efl*tan(sfa))/(efl*tan(sfa)). Failure to meet the requirements of paragraph 3.7.2.2 shall constitute failure of this tests.

4.6.7.3 Field curvature. Mount the eyepiece as the test 4.6.7.1. Mount a diopterscope so that its 5mm diameter entrance pupil remains at the goggle exit pupil while the goggle is being rotated to view a collimated target at various angles. Set the diopterscope zero diopters, and adjust the goggles for best focus on-axis. Rotate the goggles through plus and minus 18 degrees, and measure the diopters for best focus. Failure to comply with the requirements of 3.7.2.4 shall constitute failure of this test.

4.6.7.4 Transmission. Transmission of the eyepiece assembly shall be measured in accordance with MIL-STD-150, Method 6. Detector/source combination shall be in accordance with Figure 5.

4.6.7.5 Exit pupil, size and location. Using a exit pupil tester, place a back illuminated target (14.15 ± 0.15mm radius concave, 18mm diameter) in the zero diopter focal plane. With the exit pupil tester measure the size and location of the exit pupil (i.e. the image of the optical stop). Failure to meet the requirements of 3.7.2.8 shall constitute failure of this test.

4.6.7.6 Flange focal distance. The flange focal distance of the binocular eyepiece lens assembly shall be measured as part of the centered pupil MTF test, 4.6.7.8. Failure to meet the requirements of 3.4.1 shall constitute failure of this test.

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4.6.7.6.1 Flange focal distance alternate. The flange focal distance of the binocular lens group shall be measured on an optical bench utilizing a collimated light source and a measuring microscope. Perform the measurement as follows:

1. Set both eyepiece channels to "0" diopters.
2. Mount the binocular eyepiece on the optical bench so that the exit pupils are toward the collimated light source.
3. Place a 5mm aperture stop on one eye channel at the 15mm eye relief distance.
4. Place a cover over the opposite eye channel.
5. Using a measuring microscope, measure the distance from the locating flange of the collimator lens assembly to the image plane.

Failure to meet the requirements of 3.4.1 shall constitute failure of this test.

4.6.7.7 Modulation transfer function (MTF). The eyepiece shall be tested on an MTF apparatus that yields sine-wave MTF response. The MTF apparatus shall have a light source and detector whose combination shall yield a spectral response curve complying with figure (5). (If a microscope objective is used to relay the image of the eyepiece lens, the numerical aperture (NA) of the microscope objective shall not be less than 0.45). The MTF apparatus shall allow the eyepiece to be tested wither continuously or at a minimum of 4 distinct. Evenly distributed points from 0 to 40 lp/mm. The area analyzed shall be at least 0.5mm in diameter. The calibration of the MTF apparatus shall be performed before the test data is valid. The calibration procedure to be used is to measure the MTF of a known size target (e.g. slit or pinhole). All references to size are at second surface of the simulated photocathode. The MTF of a pinhole is known mathematically as first order Bessel Function. MTF = $(2J_1(x))/x$ where x= 3.14156*(lp/mm)*(pinhole diameter in mm). The MTF of a pinhole goes to zero at a spatial frequency 1.22 times its reciprocal diameter. The MTF of a slit is known mathematically as (sine (x))/(x), where x=3.14156*(lp/mm)*(slit width). The units for x is in radians. The MTF of a slit goes to zero at a spatial frequency at its reciprocal width. The procedure shall be to measure the MTF of the lens and then measure the MTF of the lens with a very large pinhole/slit. In the case of microprocessor based MTF apparatus which correct for the pinhole/slit size, the size of the pinhole/slit entered into the computer software, when measuring the lens with large pinhole/slit should not be greater than 1/40 of the larger pinhole/slit. The geometrical size of the slit shall be 0.025 ± 0.001. The MTF of the large pinhole/slit can be determined by taking the ratio of the MTF curves of the lens with large pinhole/slit, divided by the MTF of the lens under test. Differences in Modulation of greater than three percent will invalidate the MTF results. See Figure (4) for MTF as a function of spatial frequency and pinhole diameter/slit width. Failure to meet the requirements of paragraph 3.7.2.10 shall constitute failure of this test.

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4.6.7.8 Weight. The eyepiece shall be weighed on a balance or scale accurate to within ± 0.1 gram. Failure to meet the requirements of 3.7.2.11 shall constitute failure of this test.

4.6.7.9 Collimation. Setup the binocular assembly with a back illuminated target (TBD) that subtends one degree (horizontal) by one-half degree (vertical) in the zero diopter focus of the binocular assembly. View the scene through both eyepieces simultaneously.
When illuminated targets are touching each other the binocular assemblies meet the requirements of 3.7.2.12. Failure to meet the requirements of 3.7.2.11 constitutes failure of this test.

5. PACKAGING

5.1 Packaging requirements. The packaging requirements for the desired level(s) of protection shall be as specified by the acquisition activity (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The AN/PVS-7A is intended for general purpose viewing during periods of darkness.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and data of this specification.

b. That the contractor will furnish sufficient batteries used for testing.

c. Necessary actions by the contractor in the event of a Group C or D failures.

d. Sampling plans for Group B and Group C testing. As guidance, unless otherwise specified, sampling should be conducted per the requirements of MIL-STD-105.

e. Quantity and schedule for initial production testing and disposition of IPT samples for all line items.

f. Qualification - If product is not qualified at time of award the contract must require qualification prior to first delivery.

g. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).

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h. MIL-STD-810C should be used for environmental tests of section 4.

6.3 Definitions. See 3.9.

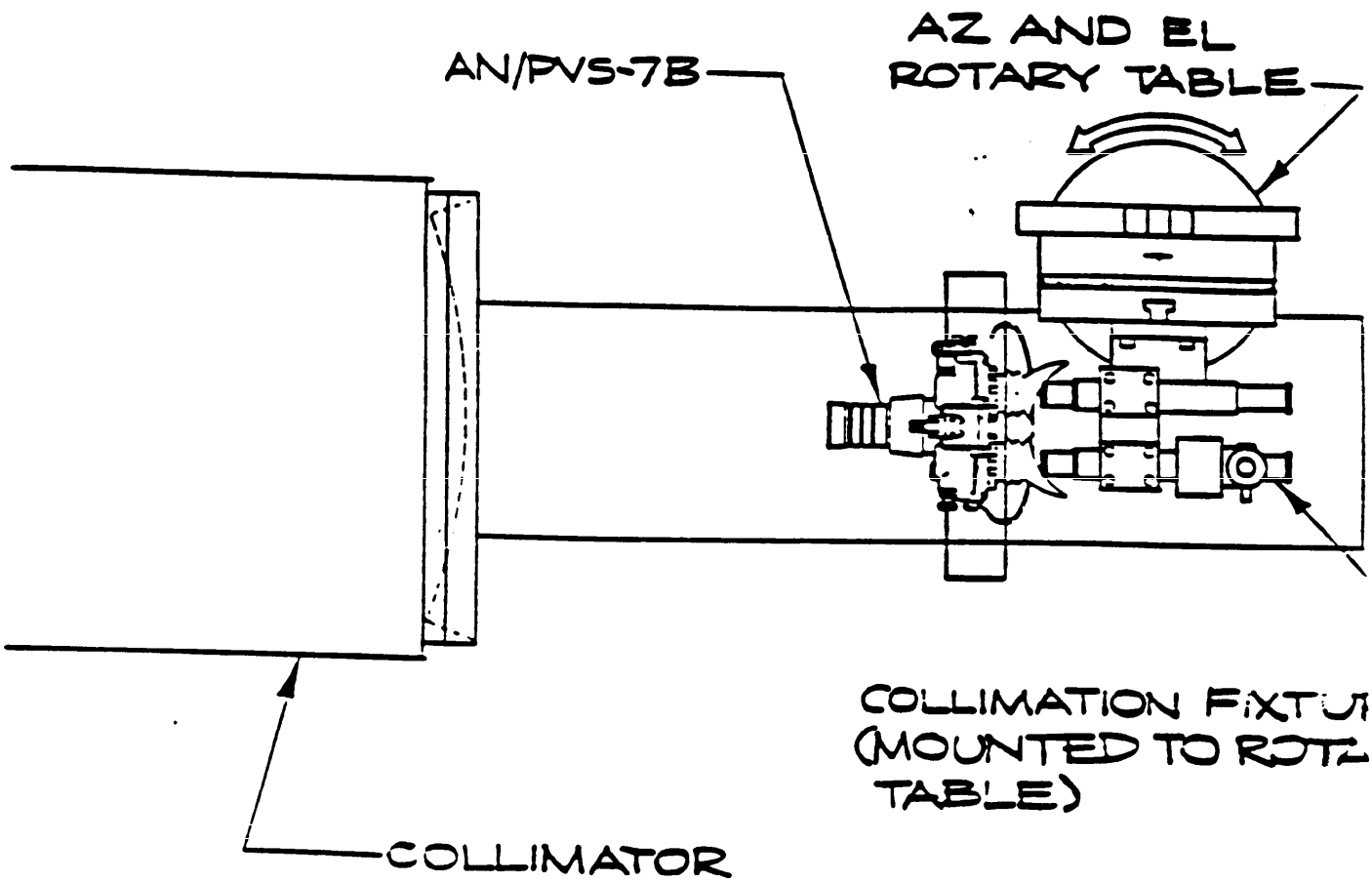
6.4 Subject term keyword listing.
Goggles
Night vision systems

Custodian
ARMY - CR

Preparing Activity
ARMY - CR

Project 5855-A321

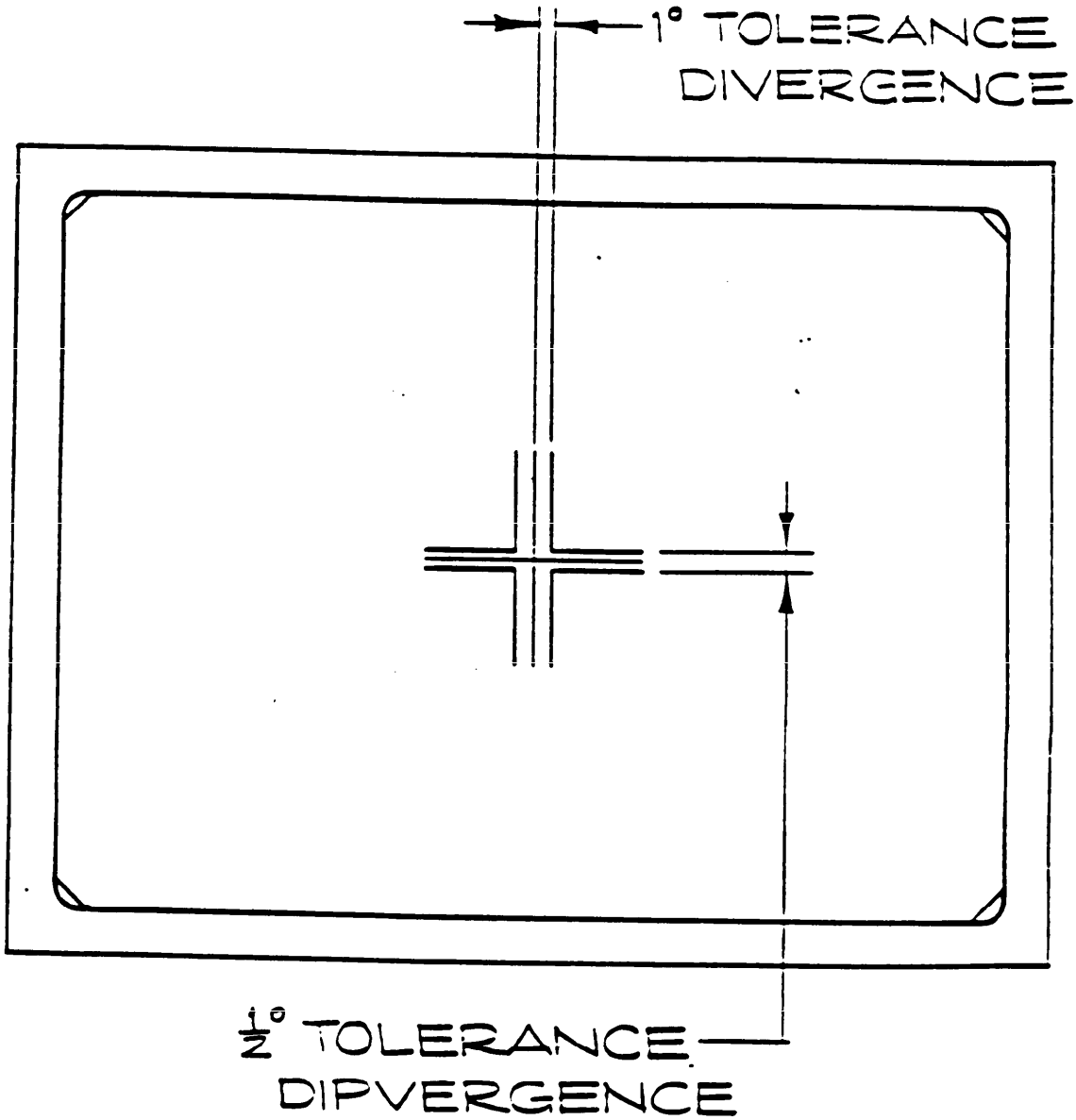
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COLLIMATION TEST SET

FIGURE 1

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TARGET, COLLIMATION

FIGURE 2

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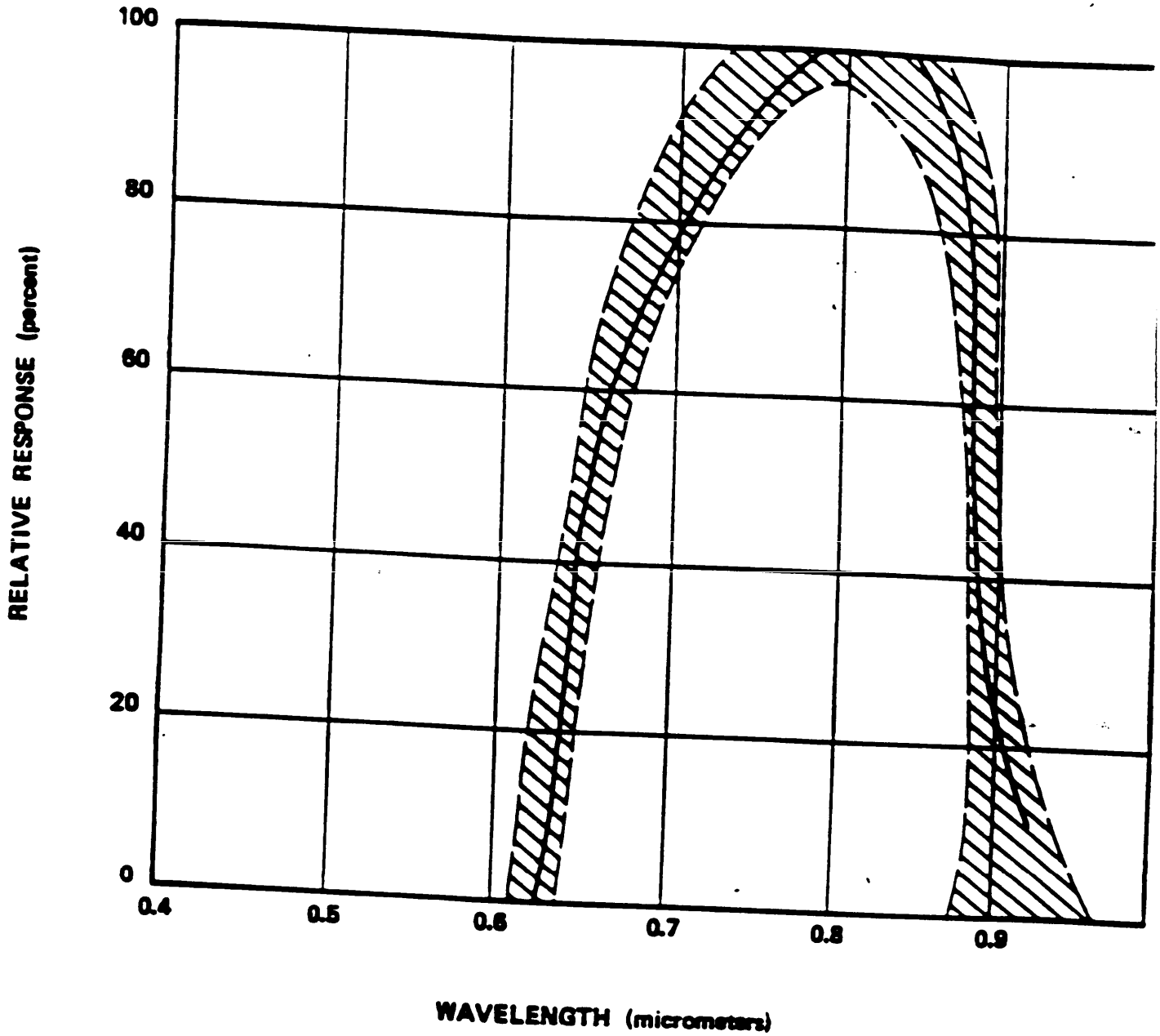


FIGURE 3. Spectral response of light source/detector combination.

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	PINHOLE	SLIT
lp/mm	MTF, %	MTF, %
0.000	100.000	100.000
2.000	99.500	99.589
4.000	98.000	98.363
6.000	95.600	96.340
8.000	93.000	93.549
10.000	89.000	90.032
12.000	84.000	85.839
14.000	78.500	81.033
16.000	73.000	75.683
18.000	67.000	69.865
20.000	60.000	63.662
22.000	54.000	57.162
24.000	46.900	50.455
26.000	40.000	43.633
28.000	33.000	36.788
30.000	26.000	30.011
32.000	19.000	23.387
34.000	13.500	17.001
36.000	8.000	10.929
38.000	4.000	5.242
40.000	0.000	0.000

PINHOLE DIAMETER - 0.0305 MM
SLIT WIDTH - 0.0250 MM

FIGURE 4 MODULATION TRANSFER FUNCTION

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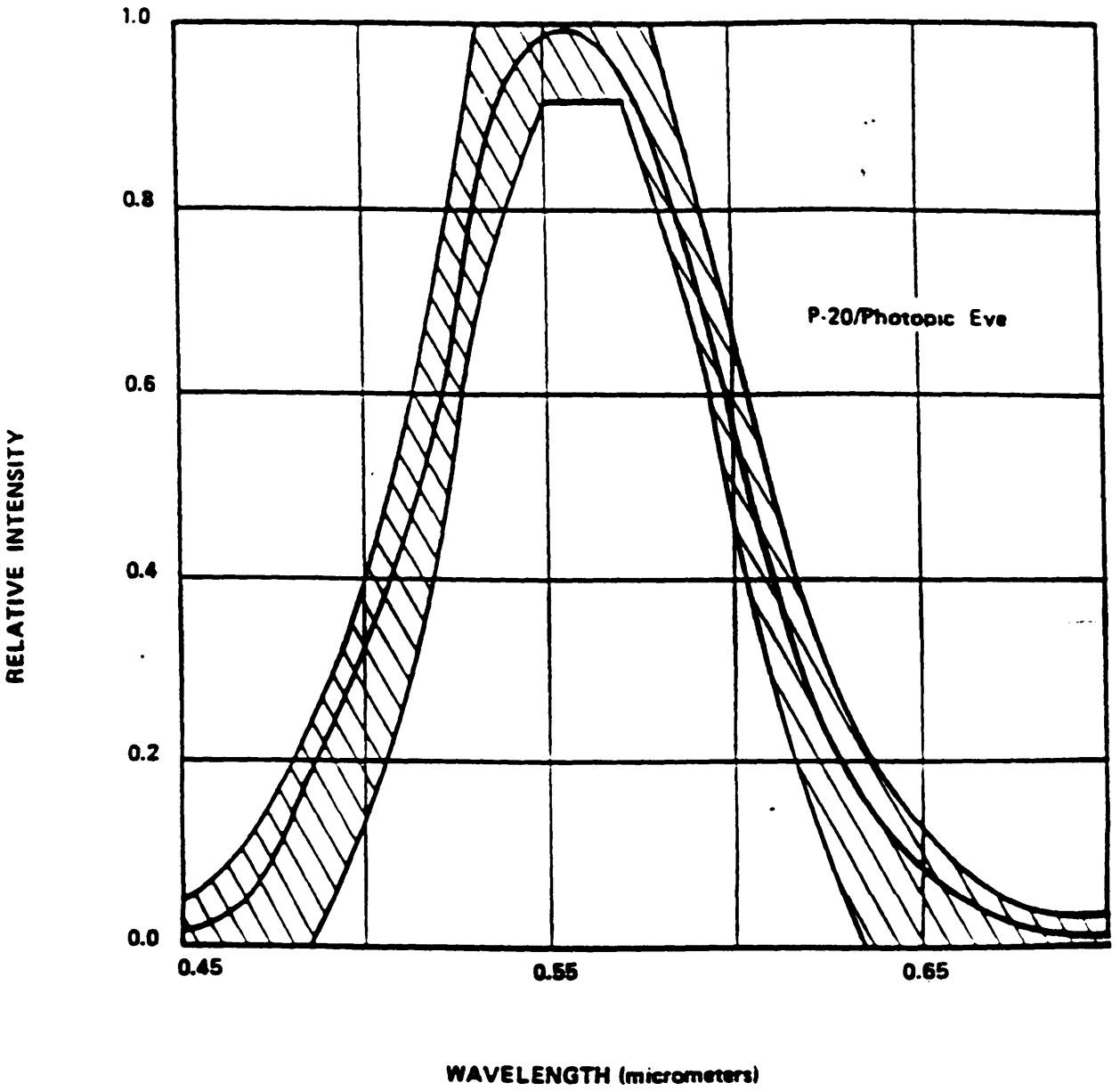
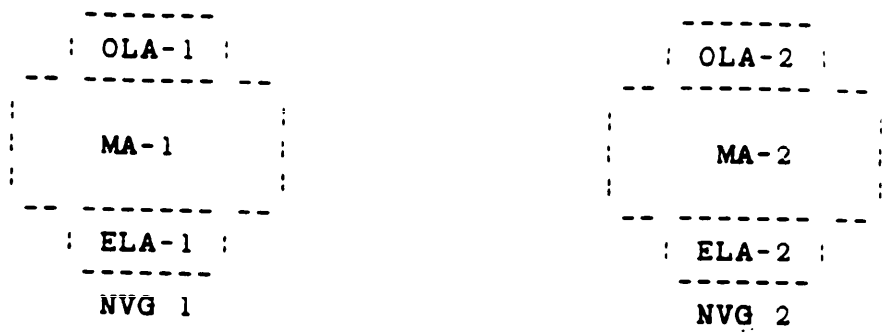


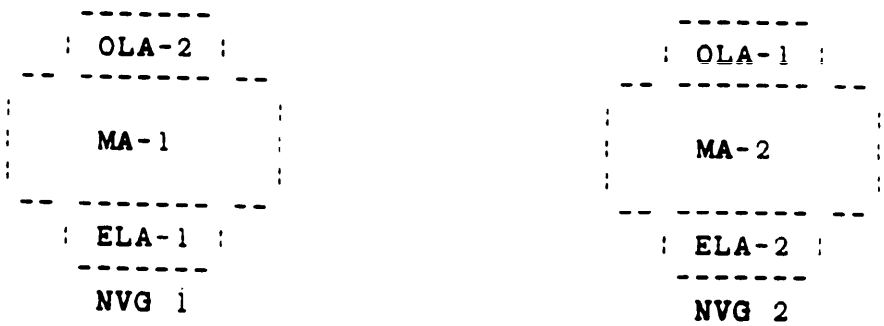
FIGURE 5. Spectal distribution of P-20/Photopic eye combination.

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PRETEST ARRANGEMENT



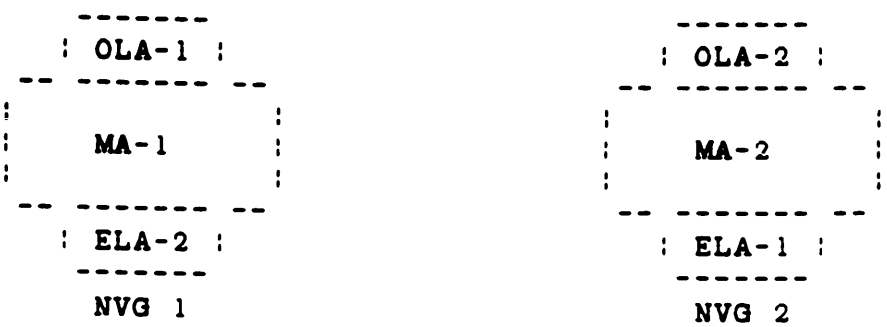
TEST ARRANGEMENT 1



TEST ARRANGEMENT 2



TEST ARRANGEMENT 3



OLA - Objective Lens Assembly
MA - Mount Assembly
ELA - Eyepiece Lens Assembly

FIGURE 6. INTERCHANGEABILITY INSPECTION ARRANGEMENTS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-49312	Goggles, Night Vision AN/PVS-7A

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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